Systems of Two Linear Equations: Intersections as Solutions
Answer Key

Grade 8 / Pre-Algebra

Quick Answers

1. $x = 3$, $y = 2$
2. $x = 4$, $y = 1$
3. $x = 3$, $y = 5$
4. $x = 2$, $y = 4$
5. $x = 2$, $y = 5$
6. $x = 2$, $y = 3$
7. $x = 6$, $y = 60$
8. —
9. $x = 3$, $y = 2$
10. —

Curriculum

Level: Grade 8 / Pre-Algebra
8.EE.C.8 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*
Difficulty 1–4 of 5 across 10 tagged checks.

1. Check $(3, 2)$ in $x + y = 5$ and $x - y = 1$.

Solution: Substitute $x = 3$, $y = 2$ into each equation separately.

$$x + y = 3 + 2 = 5 \quad \checkmark$$
$$x - y = 3 - 2 = 1 \quad \checkmark$$

Both equations are true for the same pair, so the point lies on both lines, which is exactly what it means for the graphs to cross there. The graphs intersect at $\boxed{(3,\ 2)}$.

2. Is $(4, 1)$ on $y = x - 3$ and $y = 2x - 7$?

Solution: Test the point in each equation.

$$y = x - 3: \quad 1 = 4 - 3 = 1 \quad \checkmark$$
$$y = 2x - 7: \quad 1 = 2(4) - 7 = 8 - 7 = 1 \quad \checkmark$$

Ella is right. A point that satisfies both equations is on both graphs, so it is the crossing point: $\boxed{(4,\ 1)}$.

3. Read the intersection from the table.

Solution: Scan down the two output columns and look for one row where the two y-values agree. At $x = 0, 1, 2$ the outputs are $(-1, 8)$, $(1, 7)$, $(3, 6)$ — different every time, and the gap is shrinking. At $x = 3$ both columns read 5. That single row is a point on both lines at once.

$$2(3) - 1 = 5 \quad \checkmark \qquad -(3) + 8 = 5 \quad \checkmark$$

The graphs intersect at $\boxed{(3,\ 5)}$. At $x = 4$ the columns cross over (7 versus 4), which confirms the lines have already met.

4. Second table.

Solution: The steeper line $y = 3x - 2$ starts below $y = x + 2$ and gains 2 units on it per step in x. The two columns agree only at $x = 2$, where both read 4.

$$2 + 2 = 4 \quad \checkmark \qquad 3(2) - 2 = 4 \quad \checkmark$$

Intersection: $\boxed{(2,\ 4)}$. The table shows it because that row is the one input that produces the same output for both rules.

5. Solve $y = 2x + 1$ and $y = -x + 7$ algebraically.

Solution: At the crossing point the two lines have the same y for the same x, so the two expressions for y must be equal there.

$$\begin{aligned} 2x + 1 &= -x + 7 \\ 3x &= 6 \\ x &= 2 \end{aligned}$$

Back-substitute into either equation: $y = 2(2) + 1 = 5$ (and the check $y = -2 + 7 = 5$ agrees). The solution $\boxed{(2,\ 5)}$ is the single point the two graphs share.

6. Maya's point $(8, 0)$ for $x + 2y = 8$, $3x - y = 3$.

Solution: Test her point in both equations.

$$\begin{aligned} x + 2y &= 8 + 2(0) = 8 \quad \checkmark \\ 3x - y &= 3(8) - 0 = 24 \neq 3 \quad \times \end{aligned}$$

So $(8, 0)$ is on the first line only. Satisfying one equation puts a point on one graph; the intersection has to satisfy both. Solve properly: from $3x - y = 3$ we get $y = 3x - 3$, and substituting gives

$$x + 2(3x - 3) = 8 \implies 7x - 6 = 8 \implies x = 2,$$

so $y = 3(2) - 3 = 3$. Check: $2 + 2(3) = 8$ ✓ and $3(2) - 3 = 3$ ✓. The true intersection is $\boxed{(2,\ 3)}$.

7. Break-even: $y = 5x + 30$ and $y = 10x$.

Solution: "Cost the same" means the two cost graphs have equal y at the same x — the intersection point.

$$\begin{aligned} 10x &= 5x + 30 \\ 5x &= 30 \\ x &= 6 \end{aligned}$$

Then $y = 10(6) = 60$, and Plan A agrees: $5(6) + 30 = 60$. The plans cost the same at $\boxed{(6,\ 60)}$: at 6 visits both cost 60 dollars. Below 6 visits Plan B is cheaper; above 6 visits Plan A is cheaper, which is what the crossing of the two graphs shows.

8. No intersection: $y = 2x + 1$ and $y = 2x - 4$.
Model answer (open response — grade the reasoning): Read across each row of the table: 1 versus -4, 3 versus -2, 5 versus 0, 7 versus 2. The second line's output is always exactly 5 less than the first line's output, at every x, so no input can ever make them equal. Both equations have slope 2, so the lines rise at the same rate and stay a constant vertical distance apart — they are parallel and never meet. Because a solution of the system is a point on *both* graphs and there is no such point, the system has **no solution**. Algebra says the same thing: setting $2x + 1 = 2x - 4$ gives $1 = -4$, a false statement, so no x works.
Look for: equal slopes, different y-intercepts, never meet, no solution. Full credit does not require the algebraic check.

9. Solve $2x + 3y = 12$ and $x - y = 1$.
Solution: Neither is solved for y, so isolate a variable in the simpler equation: $x - y = 1$ gives $x = y + 1$. Substitute into the first equation.

$$\begin{aligned} 2(y+1) + 3y &= 12 \\ 2y + 2 + 3y &= 12 \\ 5y &= 10 \\ y &= 2 \end{aligned}$$

Back-substitute: $x = 2 + 1 = 3$. Check both originals: $2(3) + 3(2) = 12$ ✓ and $3 - 2 = 1$ ✓. The graphs cross at $\boxed{(3,\ 2)}$.

10. Build your own system with solution $(4, -1)$.
Model answer (open response — many correct answers): One easy method is to pick any two slopes and force each line through $(4, -1)$ with point-slope form. Taking slope 1 gives $y + 1 = 1(x - 4)$, that is $y = x - 5$; taking slope -2 gives $y + 1 = -2(x - 4)$, that is $y = -2x + 7$. So

$$y = x - 5 \qquad \text{and} \qquad y = -2x + 7$$

is one valid system. Justification the student should give: substituting $x = 4$ makes the first equation read $y = -1$ and the second read $y = -8 + 7 = -1$, so the point is on both graphs, and two lines with different slopes meet in exactly one point — so $(4, -1)$ is the only solution.
Grading: accept any two distinct lines that both contain $(4, -1)$. Substitute the student's own equations at $x = 4$ and confirm each gives $y = -1$; the two slopes must differ, or the "system" is one line written twice.